

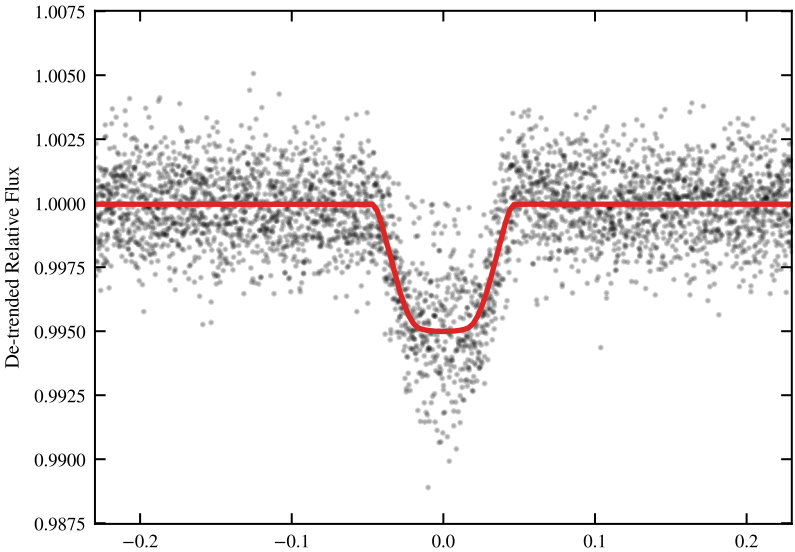
# TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 379286801 | RA: 283.30184° | Dec: -54.91081° | Sectors: 93, 94

## Derived Planetary Parameters

| Parameter  | Derived Value (MCMC) |
|------------|----------------------|
| Period (d) | 4.4531 ± 0.000122    |
| Rp/R★      | 0.073244 ± 0.00214   |
| Rp (R⊕)    | 12.7 ± 0.616         |
| b          | 0.9015 ± 0.00924     |
| T14 (h)    | 2.206 ± 0.0405       |
| a (AU)     | 0.066637 ± 0.00323   |
| Teq (K)    | 1605.8 ± 42.5        |

## Primary Phased Transit



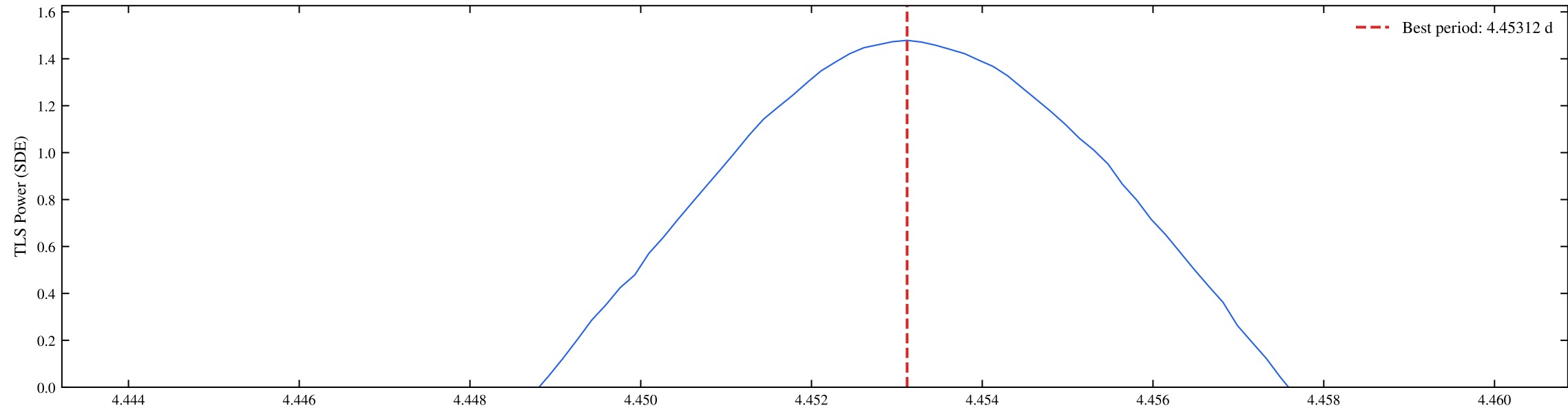
## Literature Comparison & MCMC Diagnostics

| Parameter / Metric | Literature Value | Derived / Diagnostic |
|--------------------|------------------|----------------------|
| P1 Period (d)      | 4.45312          | 4.453135 ± 0.00012   |
| P1 Epoch (BTJD)    | N/A              | 3831.3084 ± 0.0008   |
| Max R-hat (R)      | N/A              | 1.0014               |
| Min ESS            | N/A              | 4697                 |
| MCMC Divergences   | N/A              | 0                    |

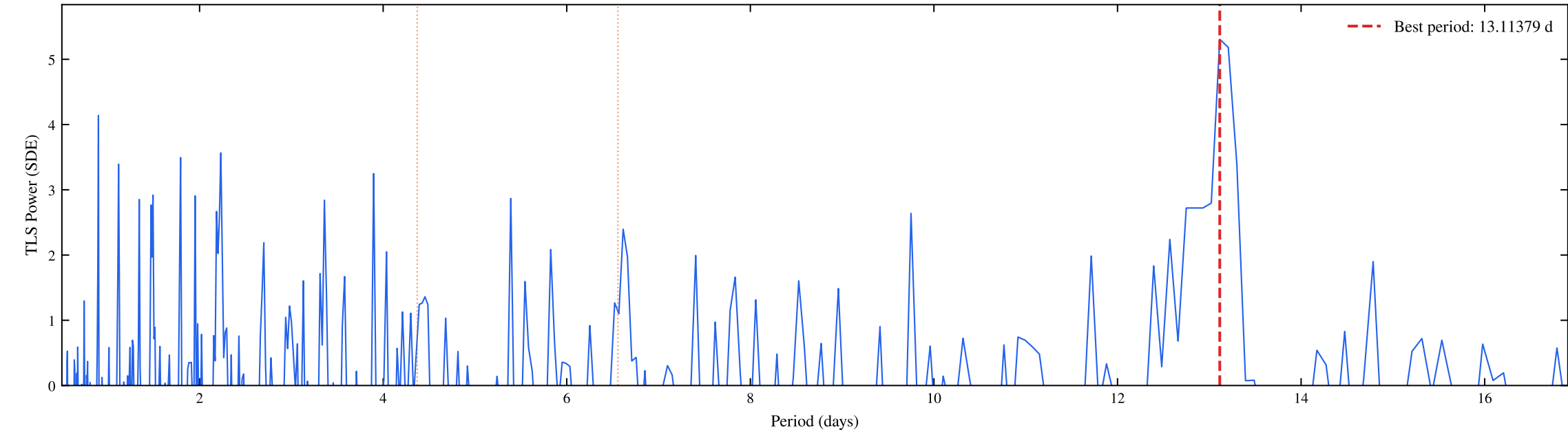
## Host Star Properties

| Stellar Parameter | Value           | Source      |
|-------------------|-----------------|-------------|
| R★ (R☉)           | 1.586 ± 0.061   | VizieR TIC8 |
| M★ (M☉)           | 1.47 ± 0.202    | VizieR TIC8 |
| Teff (K)          | 6829 ± 154      | VizieR TIC8 |
| log g (dex)       | 4.205 ± 0.0771  | VizieR TIC8 |
| [Fe/H]            | N/A             | Catalog     |
| ρ★ (g/cm³)        | 0.5195 ± 0.0932 | Derived     |

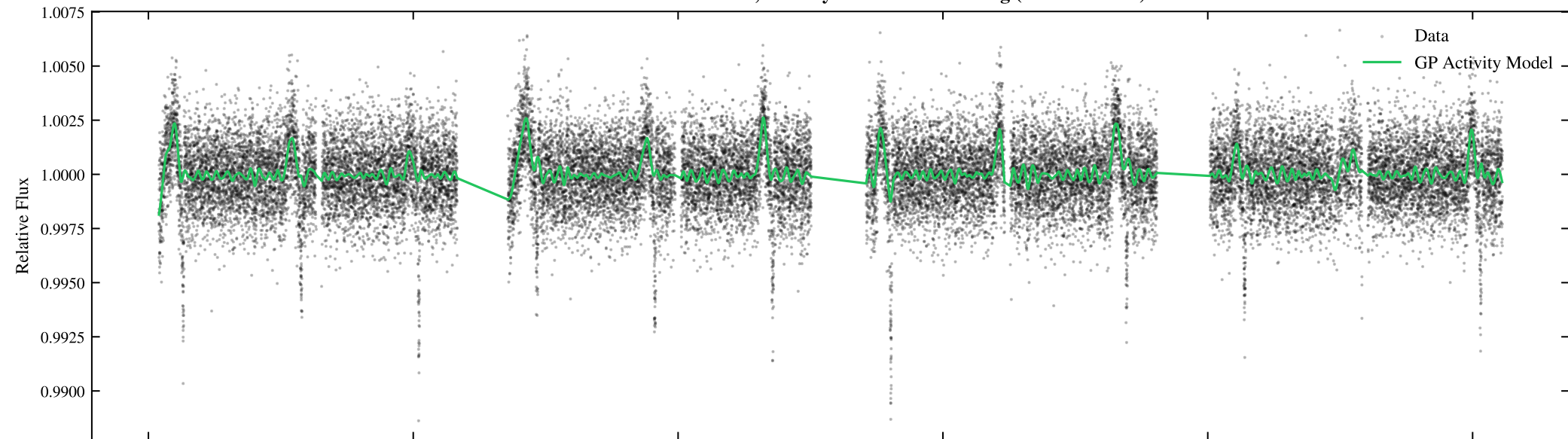
TIC 379286801 | Sectors: 93, 94 | TLS Periodogram (Planet 1 Search) (SDE = 1.48)



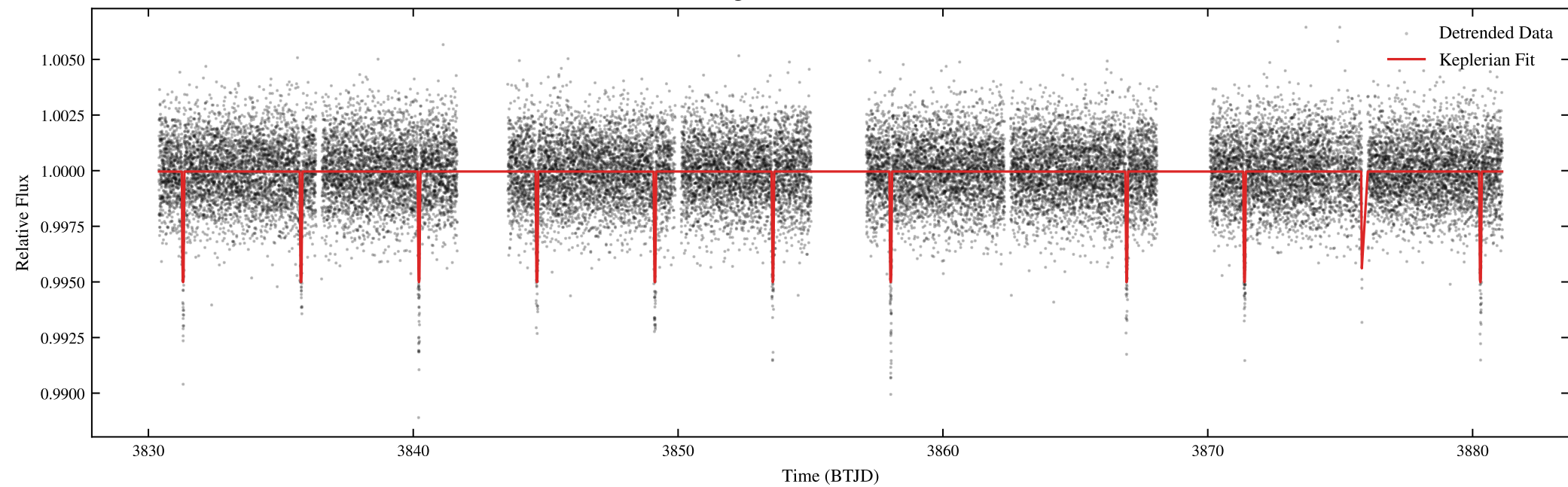
TIC 379286801 | Sectors: 93, 94 | TLS Periodogram (Planet 2 Search) (SDE = 5.31)



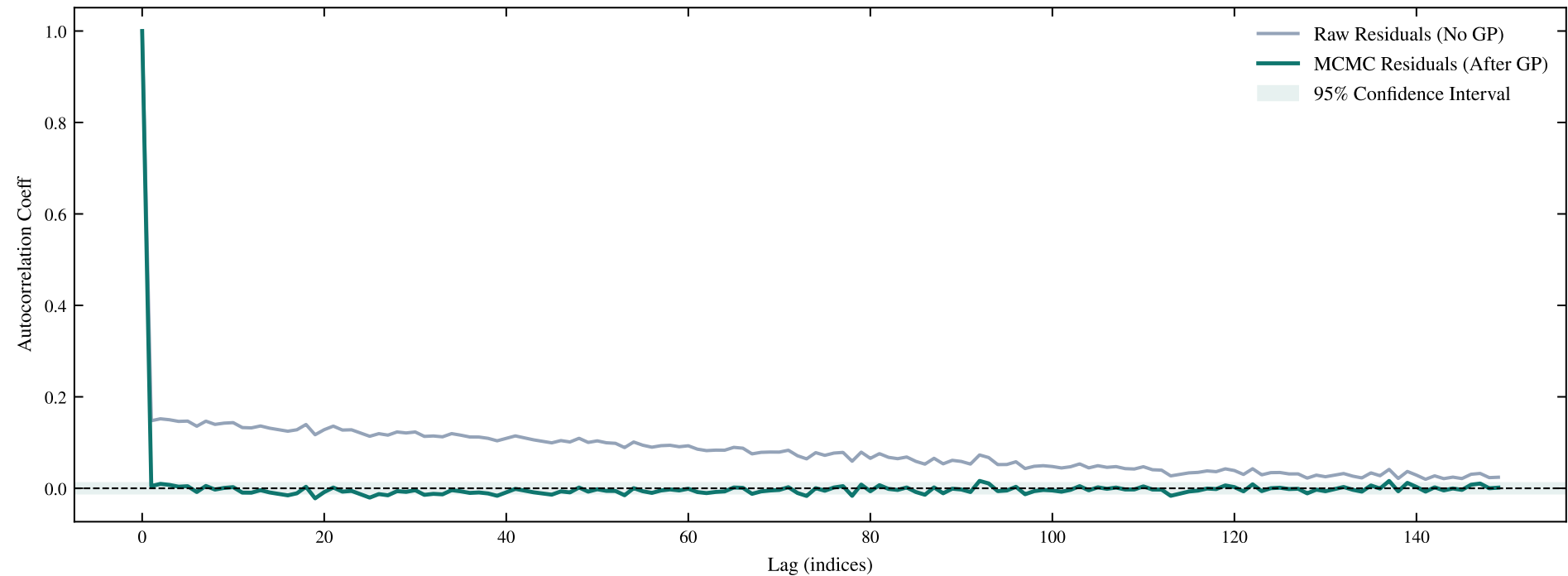
TIC 379286801 | Sectors: 93, 94 | Bayesian GP Detrending (Time Domain)



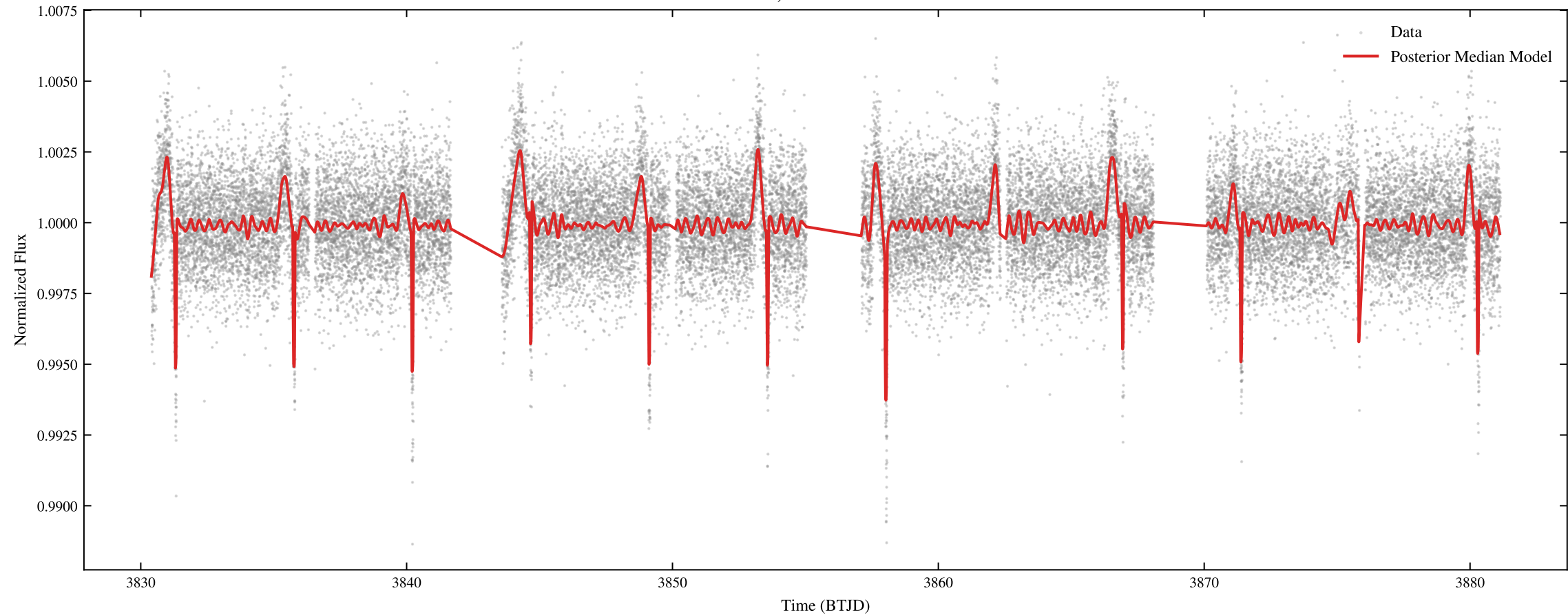
Detrended Light Curve & Best-Fit Transit Model

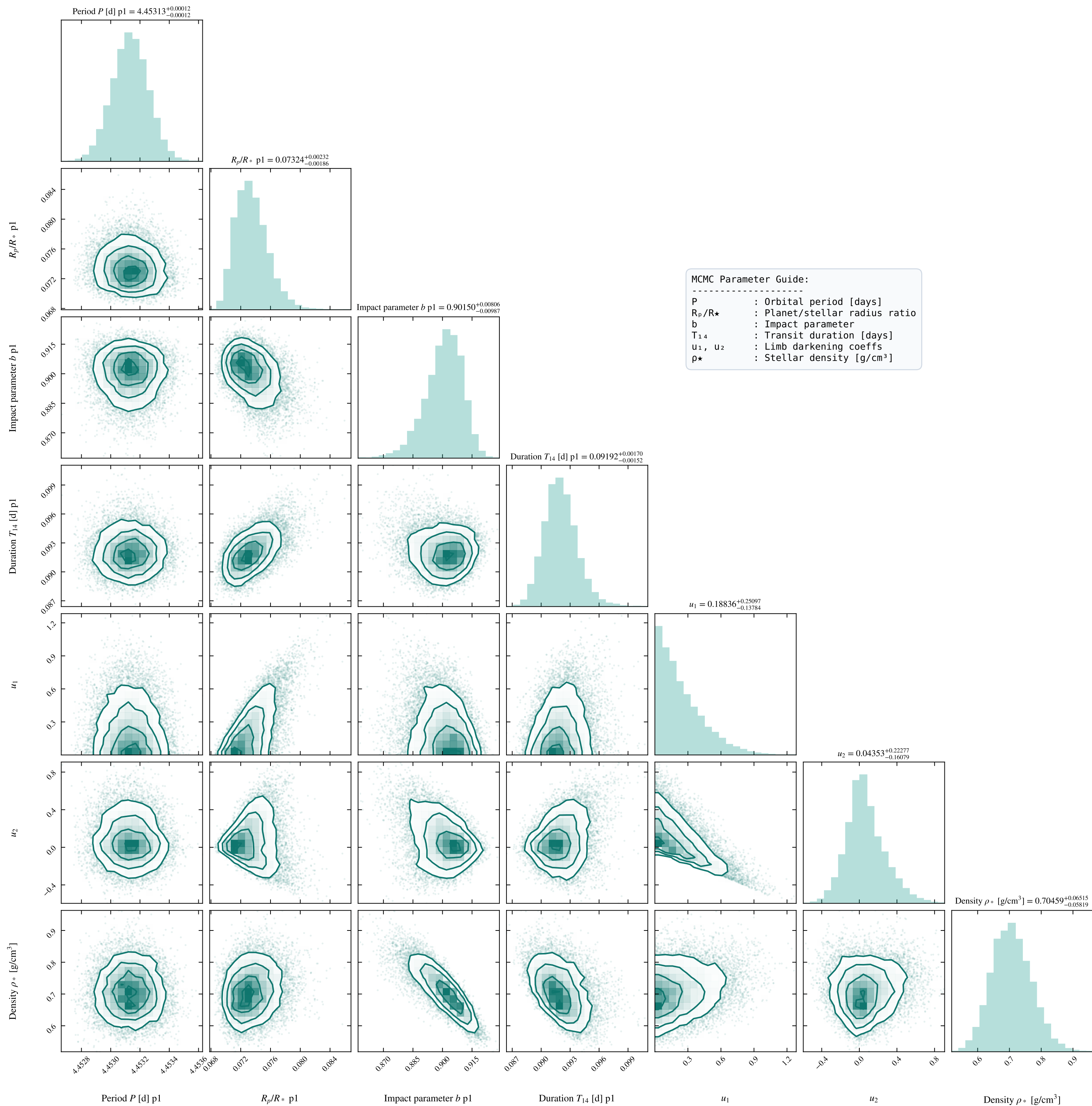


TIC 379286801 | Sectors: 93, 94 | GP Residuals Autocorrelation Function (ACF)

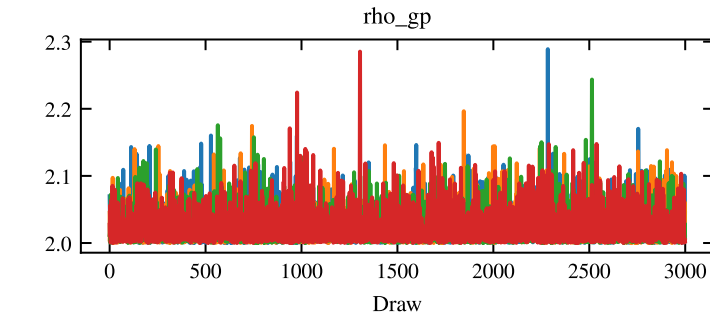
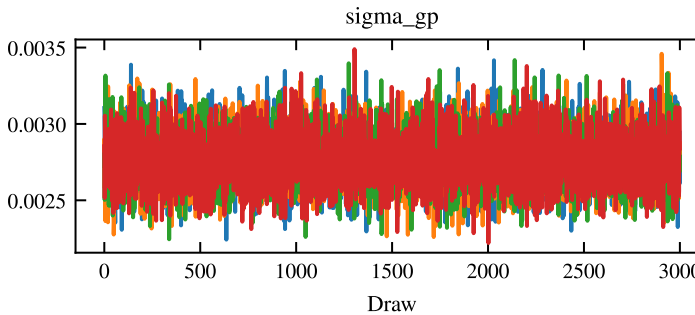
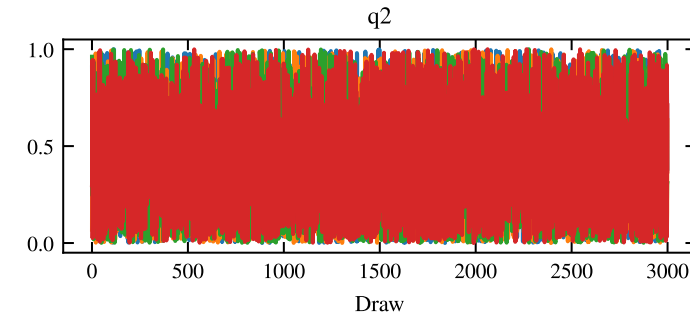
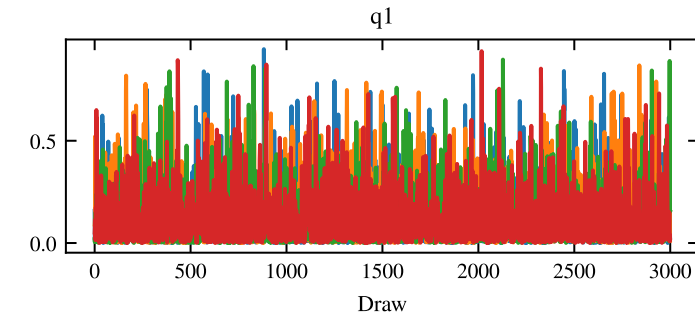
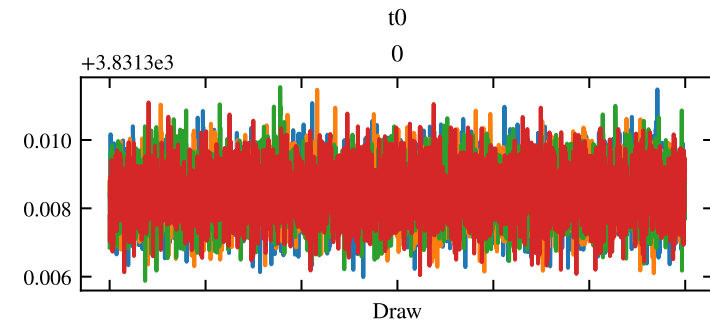
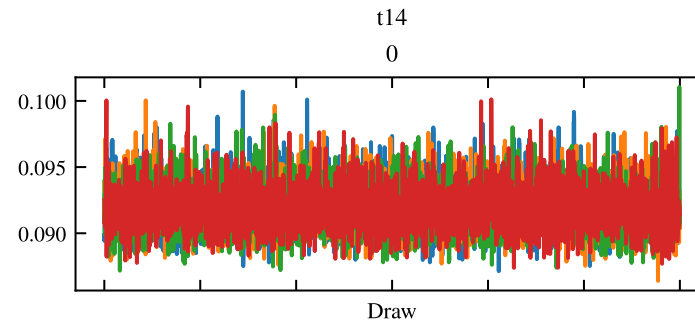
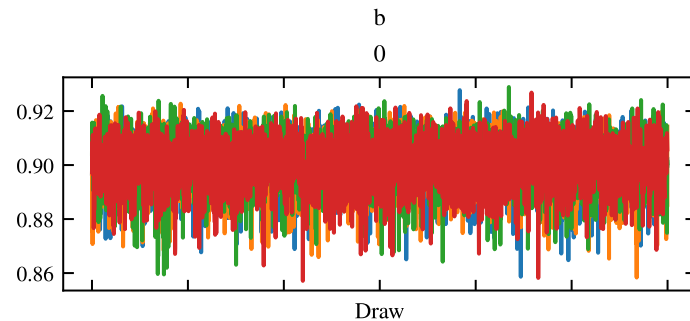
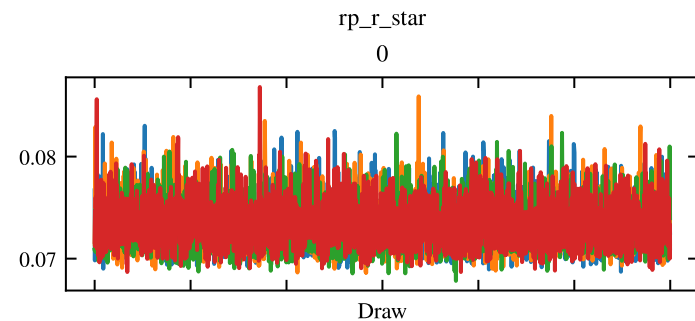


TIC 379286801 | Sectors: 93, 94 | MCMC Posterior Predictive Check

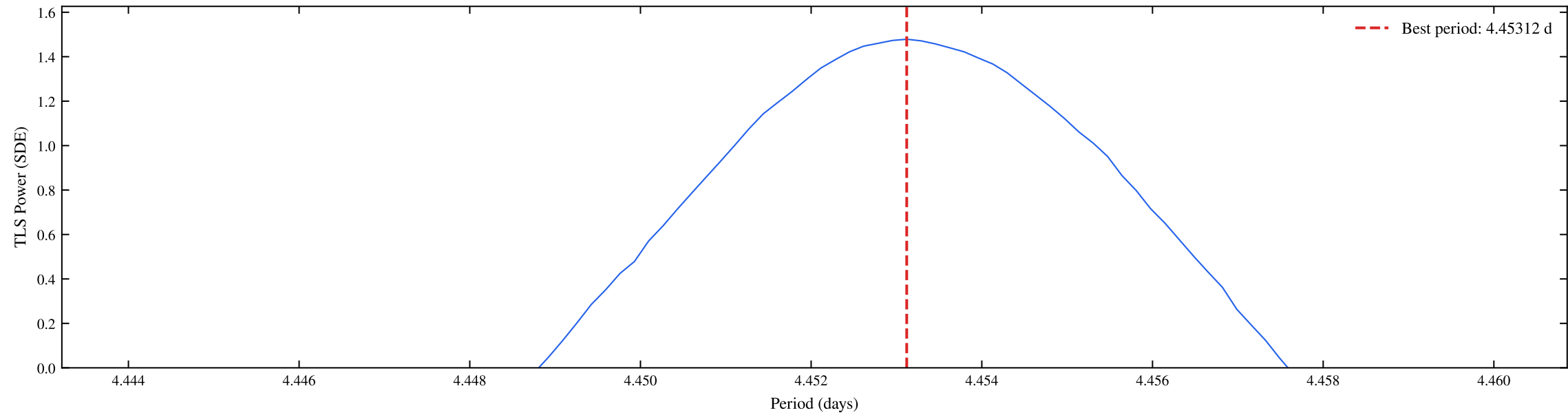




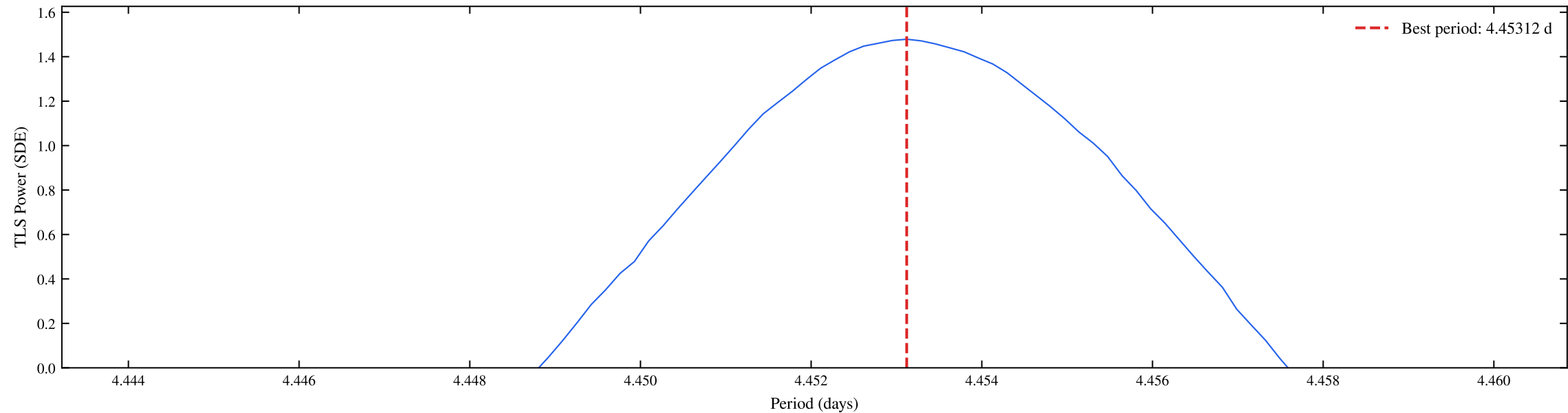
# TIC 379286801 | Sectors: 93, 94 | MCMC Trace Plots



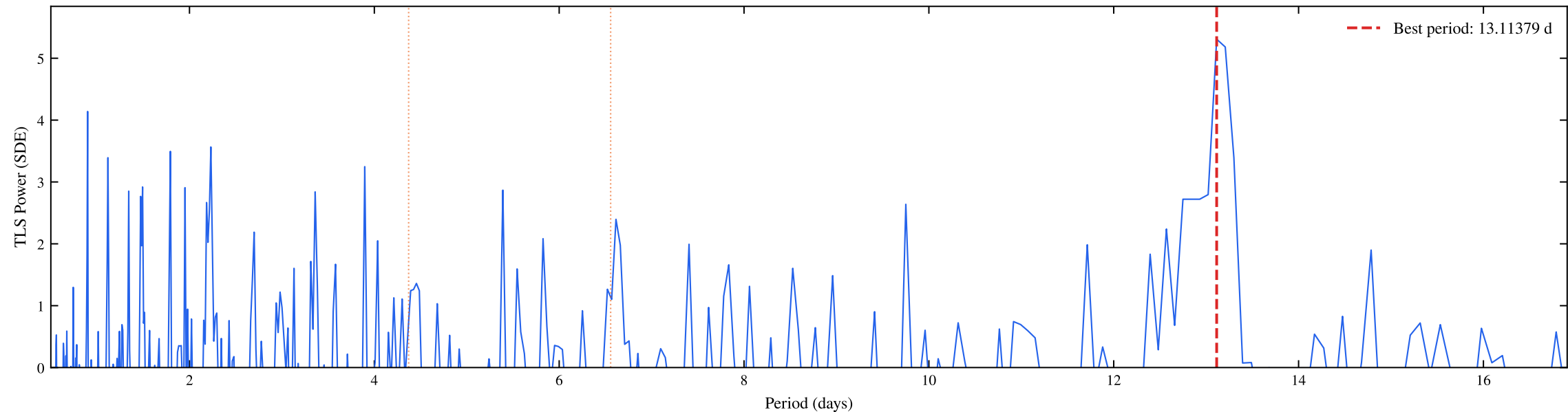
TIC 379286801 | Sectors: 93, 94 | TLS Coarse Periodogram (SDE = 1.48)



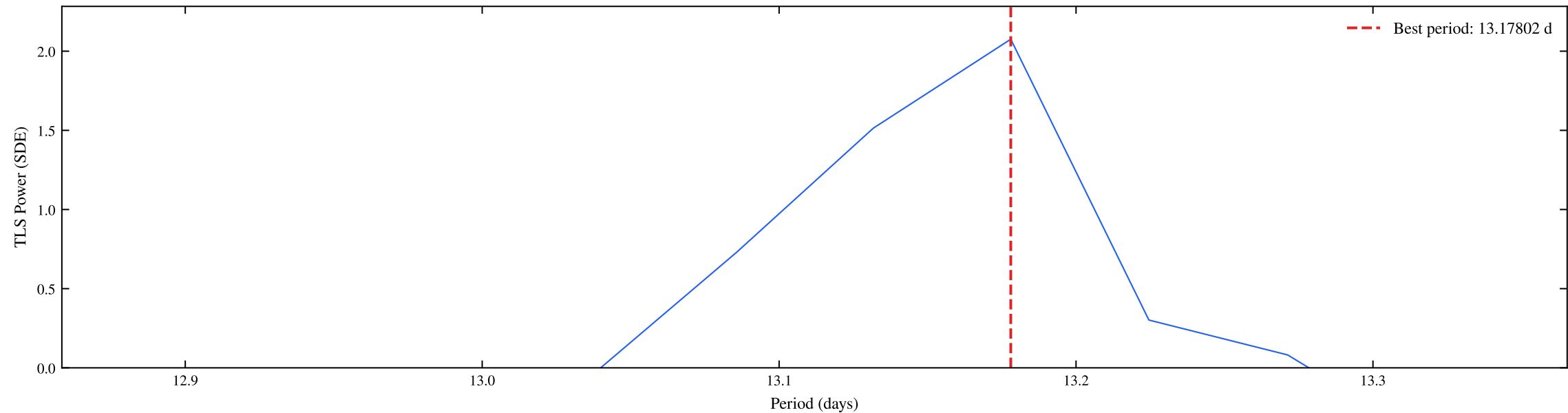
TIC 379286801 | Sectors: 93, 94 | TLS Fine Periodogram (SDE = 1.48)



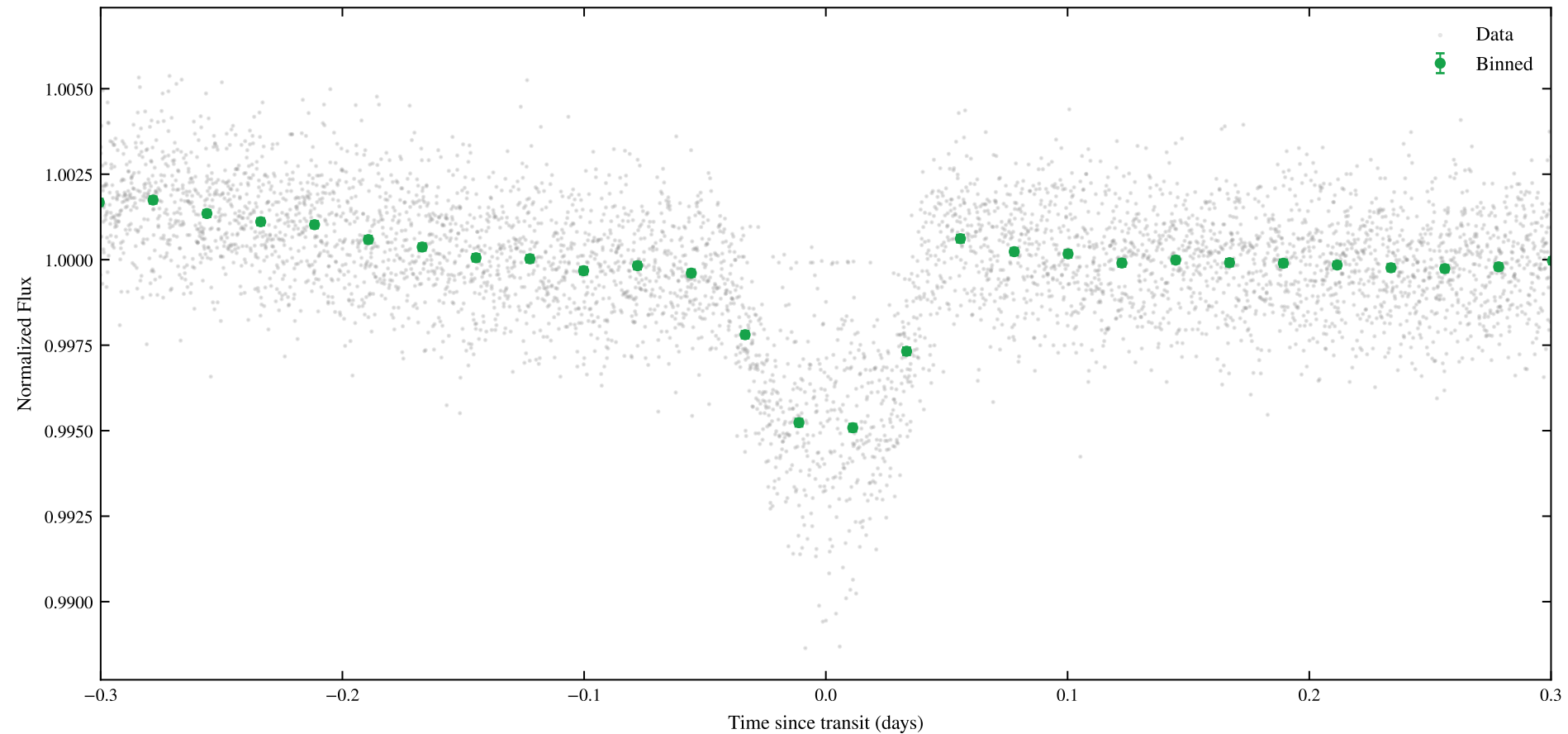
TIC 379286801 | Sectors: 93, 94 | TLS Coarse Periodogram (SDE = 5.31)



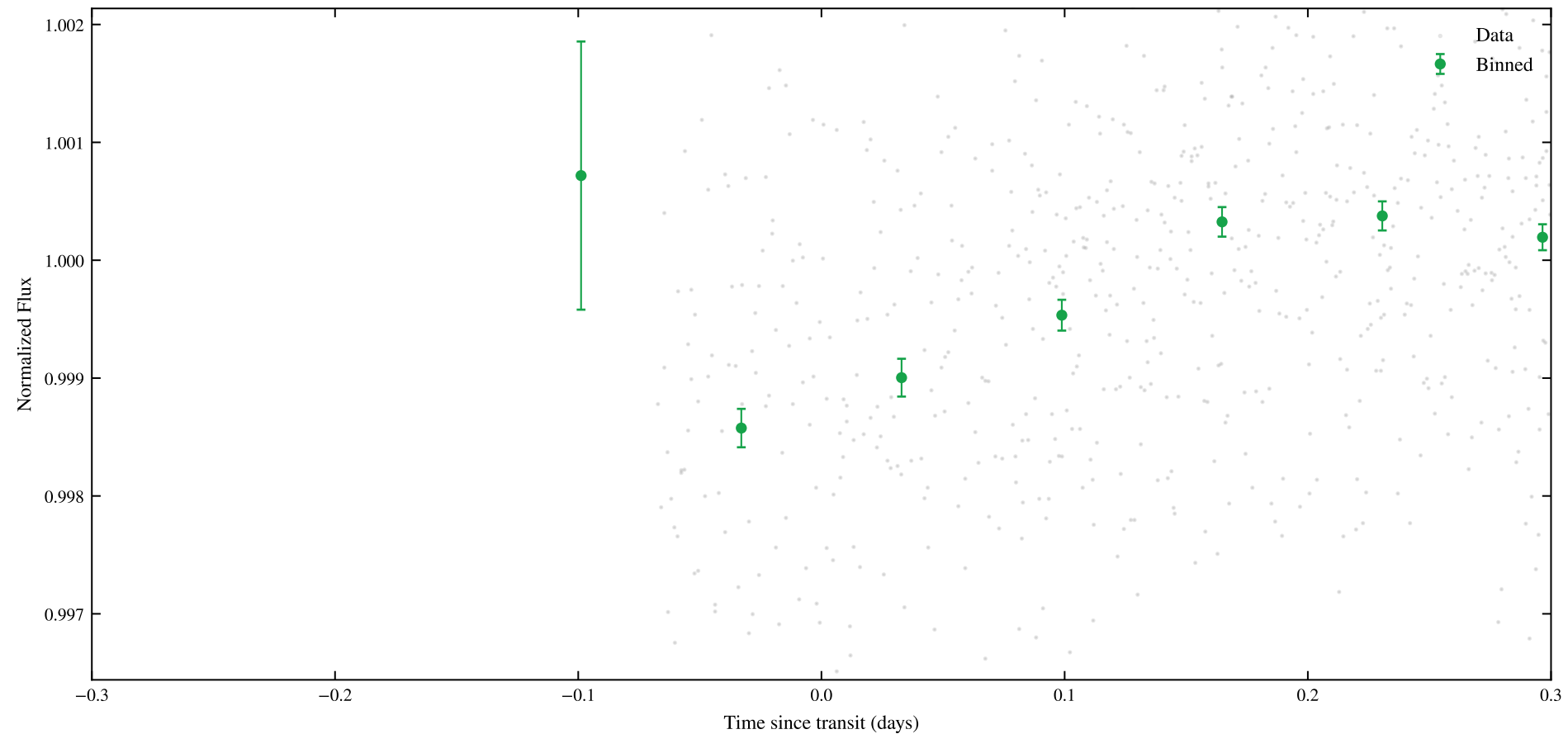
TIC 379286801 | Sectors: 93, 94 | TLS Fine Periodogram (SDE = 2.08)



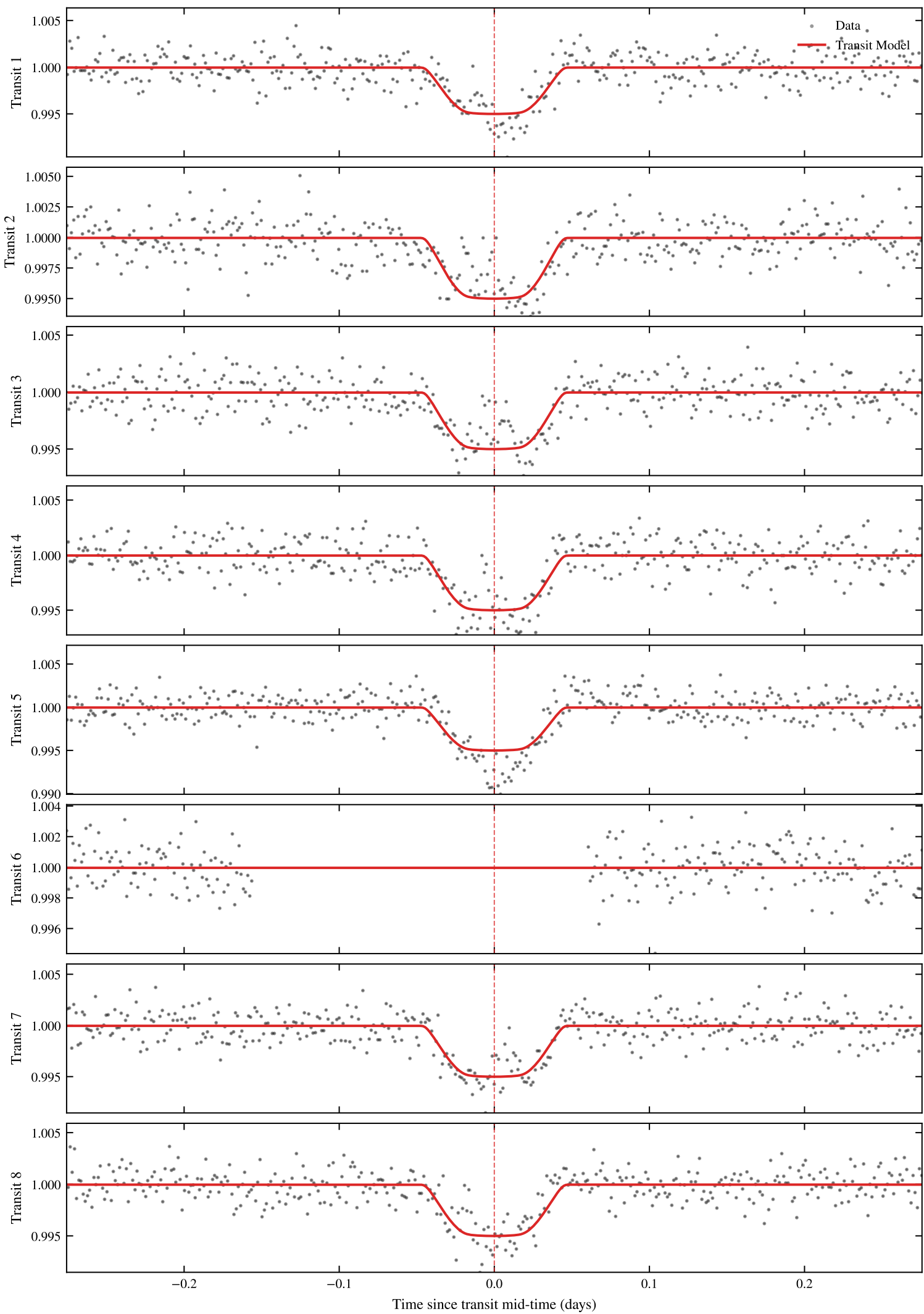
TIC 379286801 | Sectors: 93, 94 | Phase-Folded Transit (P = 4.45312 d)



TIC 379286801 | Sectors: 93, 94 | Phase-Folded Transit (P = 13.17802 d)



# TIC 379286801 | Sectors: 93, 94 | Stacked Individual Transits



TIC 379286801 | Sectors: 93, 94 | Phased MCMC Fit - Planet 1

